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Quantum Spring

Reinhold Bertlmann starts each day with a tiny act of rebellion. He doesn't look like a rebel at first glance—his impeccably trimmed facial hair and his professorial taste in clothes match the formal style of his hometown, Vienna, which has never really shed its imperial facade. But Bertlmann's sartorial conformity stops just short of his shoes: his socks are always mismatched. "I've worn socks of different colors since my early student days. And I am a student of the so-called '68 generation," Bertlmann says. "And this was my little protest. My hidden protest. To wear socks of different colors, because I realized that whenever somebody sees this, they were either shocked—they said, 'how stupid, how can you do it?'—or they laughed about it and thought I am crazy."

Forty years ago, Bertlmann's rebellion was more obvious. With shoulder-length hair and an unruly beard, he stuck out when he first arrived at CERN in 1978. "An American would say [I was] a hippie or something," he recalled. Nonetheless, Bertlmann's open, friendly smile attracted many friends at CERN, and most eventually noticed his socks. But John Bell never mentioned them. Bertlmann and Bell worked together for two years on a thorny calculation in particle physics, totally unrelated to Bell's theorem. "He did not say one word [about my socks], not one word," Bertlmann recalled. And Bertlmann, in turn, did not ask Bell about the rumor he had heard in the CERN canteen: that Bell had done some kind of important work in the foundations of quantum physics. "People said, 'Oh, you're collaborating with Bell? He is somehow famous in quantum